# Math 401 — Applications of Linear Algebra

Course Notes

Least Squares · Curve Fitting · Team Ranking · Graph Theory · Markov Chains · PageRank · SVD · Image Compression · Quantum Qubit

Based on notes by Justin Wyss-Gallifent
and Lay & McDonald, *Linear Algebra and Its Applications*, Ch. 10

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# Chapter 1

# Linear Algebra Review

This chapter summarizes the prerequisite linear algebra needed for this course. Source: Justin Wyss-Gallifent, *Math 401/431: Linear Algebra Review.*

## 1.1 Matrices and Vectors

**Definition 1.1.1** (Matrix)**.** A **matrix** is a rectangular array of numbers. An $n \times m$ matrix has $n$ rows and $m$ columns. A **vector** is an $n \times 1$ matrix.

**Definition 1.1.2** (Matrix–Vector Product)**.** If $A$ is $n \times m$ and $\mathbf{b}$ is $m \times 1$, the product $A\mathbf{b}$ is the linear combination of columns of $A$ weighted by the entries of $\mathbf{b}$:

$$A\mathbf{b} = b_1\mathbf{a}_1 + \cdots + b_m\mathbf{a}_m.$$

**Definition 1.1.3** (Transpose, Symmetry, Inverse)**.**

- The **transpose** $A^T$ is obtained by swapping rows and columns: $(A^T)_{ij} = A_{ji}$.
- $A$ is **symmetric** if $A^T = A$.
- A square matrix $A$ is **invertible** if there exists $A^{-1}$ with $AA^{-1} = A^{-1}A = I$.

**Theorem 1.1.1** ($2 \times 2$ Inverse)**.** $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ *provided* $ad - bc \neq 0$.

**Definition 1.1.4** (Dot Product and Norm)**.** For $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T\mathbf{w} = v_1w_1 + \cdots + v_nw_n, \qquad \|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}.$$

## 1.2 Determinants

**Definition 1.2.1** (Determinant)**.** For a $2 \times 2$ matrix: $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. For larger matrices, expand along the first row:

$$\det(A) = a_{11}\det(A_{11}) - a_{12}\det(A_{12}) + \cdots \pm a_{1n}\det(A_{1n}),$$

where $A_{1j}$ is the matrix with row 1 and column $j$ removed.

**Theorem 1.2.1.** *A square matrix $A$ is invertible if and only if* $\det(A) \neq 0$.

## 1.3 Systems of Equations

**Theorem 1.3.1.** *Any linear system $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$ can be written as $A\mathbf{x} = \mathbf{b}$. It has:*

- *Exactly one solution if $A$ is invertible (then $\mathbf{x} = A^{-1}\mathbf{b}$).*
- *No solution or infinitely many solutions otherwise.*

## 1.4 Linear Independence

**Definition 1.4.1.** A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is **linearly independent** if $a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$ implies $a_1 = \cdots = a_n = 0$. Otherwise it is **linearly dependent** (one vector is a linear combination of the others).

## 1.5 Vector Spaces and Bases

**Definition 1.5.1** (Vector Space)**.** A nonempty set $V$ of vectors is a **vector space** (subspace of $\mathbb{R}^n$) if it contains $\mathbf{0}$, is closed under addition, and is closed under scalar multiplication.

**Definition 1.5.2** (Span, Basis, Column Space)**.**

- $\text{span}(S) = \{a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k \mid a_i \in \mathbb{R}\}$ for $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$.
- A **basis** for $V$ is a linearly independent set $B$ with $\text{span}(B) = V$.
- The **column space** of $A$ is $\text{Col}(A) = \text{span}(\text{columns of } A) = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^m\}$.
- The **dimension** $\dim(V)$ is the number of vectors in any basis of $V$.

## 1.6 Orthogonality and Orthonormality

**Definition 1.6.1** (Orthogonal and Orthonormal Sets)**.**

- $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is **orthogonal** if $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for $i \neq j$.
- It is **orthonormal** if additionally $\|\mathbf{v}_i\| = 1$ for all $i$.

**Definition 1.6.2** (Orthogonal Matrix)**.** A square matrix is **orthogonal** if its columns form an orthonormal set, equivalently $A^TA = AA^T = I$, i.e., $A^{-1} = A^T$.

*Remark* 1.6.1. Orthogonal matrices are computationally convenient because $A^{-1} = A^T$ is trivially obtained. They arise naturally in QR decompositions used for least-squares computation.

## 1.7 Eigenvalues and Eigenvectors

**Definition 1.7.1** (Eigenvalue and Eigenvector)**.** If $A$ is $n \times n$, then $\lambda$ is an **eigenvalue** and $\mathbf{v} \neq \mathbf{0}$ is the corresponding **eigenvector** if $A\mathbf{v} = \lambda\mathbf{v}$.

**Definition 1.7.2** (Characteristic Polynomial)**.** $\text{char}(A) = \det(\lambda I - A)$. The eigenvalues of $A$ are the roots of $\text{char}(A)$.

**Fact 1.7.1.** *For a $2 \times 2$ matrix:* $\text{char}(A) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$. *For a triangular matrix: eigenvalues are the diagonal entries.*

**Example 1.7.1.** $A = \begin{pmatrix} 4 & 7 \\ 1 & -2 \end{pmatrix}$. Then $\text{char}(A) = \lambda^2 - 2\lambda - 15 = (\lambda - 5)(\lambda + 3)$. Eigenvalues: $\lambda_1 = 5$ with eigenvector $\begin{pmatrix} 7 \\ 1 \end{pmatrix}$; $\lambda_2 = -3$ with eigenvector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

**Theorem 1.7.1.** *The eigenspace for eigenvalue $\lambda$ (the set of all eigenvectors plus $\mathbf{0}$) is a vector space with dimension $\leq$ multiplicity of $\lambda$ as a root of* $\text{char}(A)$.

## 1.8 Diagonalizable Matrices

**Definition 1.8.1.** $A$ is **diagonalizable** if $A = PDP^{-1}$ for invertible $P$ and diagonal $D$. The columns of $P$ are eigenvectors and the diagonal of $D$ contains corresponding eigenvalues.

**Theorem 1.8.1.** *$A$ is diagonalizable $\iff$ the dimension of each eigenspace equals the multiplicity of the eigenvalue.*

**Theorem 1.8.2** (Orthogonal Diagonalization)**.** *$A$ is **orthogonally diagonalizable** (i.e., $A = QDQ^T$ with $Q$ orthogonal) if and only if $A$ is symmetric.*

*Remark* 1.8.1. Diagonalization is useful for computing powers: $A^k = PD^kP^{-1}$. Since $D$ is diagonal, $D^k = \text{diag}(\lambda_1^k, \ldots, \lambda_n^k)$ is easy to compute. This is fundamental to analyzing Markov chain convergence.

## 1.9 Linear Transformations

**Definition 1.9.1.** $T : \mathbb{R}^n \to \mathbb{R}^m$ is **linear** if $T(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha T(\mathbf{v}) + \beta T(\mathbf{w})$ for all scalars $\alpha, \beta$ and vectors $\mathbf{v}, \mathbf{w}$.

**Theorem 1.9.1.** *Every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is represented by an $m \times n$ matrix $A = [T(\mathbf{e}_1) \ \cdots \ T(\mathbf{e}_n)]$, so $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v}$.*

**Example 1.9.1.** $T(x_1, x_2) = (2x_1,\ x_2 - x_1,\ 3x_1)$ is linear with matrix $A = \begin{pmatrix} 2 & 0 \\ -1 & 1 \\ 3 & 0 \end{pmatrix}$.

Note: $T(\mathbf{0}) = \mathbf{0}$ is a necessary condition for linearity. If $T(\mathbf{0}) \neq \mathbf{0}$ (e.g., $T(x) = x + 2$), the transformation is not linear.

# Chapter 2

# Least Squares

## 2.1 Motivation

We study the matrix equation $A\mathbf{x} = \mathbf{b}$. When no exact solution exists (i.e., $\mathbf{b} \notin \mathrm{Col}(A)$), we seek the best approximate solution – the one that makes $A\hat{\mathbf{x}}$ as close as possible to $\mathbf{b}$.

## 2.2 The Normal Equation

### Geometric Idea

The vector in $\mathrm{Col}(A)$ closest to $\mathbf{b}$ is the orthogonal projection $\hat{\mathbf{b}} = \mathrm{Pr}_{\mathrm{Col}(A)}\,\mathbf{b}$. We then seek $\hat{\mathbf{x}}$ such that $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$.

### Derivation

Since $\hat{\mathbf{b}} - \mathbf{b} \perp \mathrm{Col}(A)$, for every vector $A\bar{x} \in \mathrm{Col}(A)$:

$$(A\bar{x}) \cdot (A\hat{\mathbf{x}} - \mathbf{b}) = 0 \quad \text{for all } \bar{x} \implies A^T(A\hat{\mathbf{x}} - \mathbf{b}) = \mathbf{0}.$$

This gives the **normal equation**:

$$\boxed{A^T A\hat{\mathbf{x}} = A^T\mathbf{b}.}$$

### Pythagorean Justification (Why This Minimizes Error)

For any $\bar{x}$, write $A\bar{x} - \mathbf{b} = (A\hat{\mathbf{x}} - \mathbf{b}) + A(\bar{x} - \hat{\mathbf{x}})$. Since these two terms are perpendicular:

$$\|A\bar{x} - \mathbf{b}\|^2 = \|A\hat{\mathbf{x}} - \mathbf{b}\|^2 + \|A(\bar{x} - \hat{\mathbf{x}})\|^2 \geq \|A\hat{\mathbf{x}} - \mathbf{b}\|^2.$$

## 2.3 Definitions and Main Theorem

**Definition 2.3.1** (Least-Squares Solution)**.** A **least-squares solution** to $A\mathbf{x} = \mathbf{b}$ is any $\hat{\mathbf{x}}$ satisfying $\|A\hat{\mathbf{x}} - \mathbf{b}\| \leq \|A\bar{x} - \mathbf{b}\|$ for all $\bar{x}$.

**Definition 2.3.2** (Least-Squares Error)**.** The **least-squares error** is $\|A\hat{\mathbf{x}} - \mathbf{b}\|$.

**Theorem 2.3.1** (Uniqueness)**.**

- *If the columns of $A$ are **linearly independent**, the least-squares solution is unique: $\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b}$.*
- *If the columns are **linearly dependent**, there are infinitely many least-squares solutions.*

**Corollary 2.3.2.** *If $A$ is invertible, then $(A^T A)^{-1} A^T\mathbf{b} = A^{-1}(A^T)^{-1} A^T\mathbf{b} = A^{-1}\mathbf{b}$, i.e., least squares recovers the exact solution.*

**Fact 2.3.1.** *The normal equation always has at least one solution (since $A^T\mathbf{b} \in \mathrm{Col}(A^TA)$). This is why a least-squares solution always exists.*

## 2.4 Method: Orthogonal Projection via Gram-Schmidt

An alternative to the normal equation is to find an orthogonal basis for $\mathrm{Col}(A)$ using Gram-Schmidt, then project $\mathbf{b}$ directly:

**Gram-Schmidt:** Given linearly independent $\mathbf{c}_1, \mathbf{c}_2$:

$$\mathbf{c}_2^\perp = \mathbf{c}_2 - \frac{\mathbf{c}_1 \cdot \mathbf{c}_2}{\mathbf{c}_1 \cdot \mathbf{c}_1}\mathbf{c}_1.$$

**Projection:**

$$\mathrm{Pr}_{\mathrm{Col}(A)}\,\mathbf{b} = \frac{\mathbf{b}\cdot\mathbf{c}_1}{\|\mathbf{c}_1\|^2}\mathbf{c}_1 + \frac{\mathbf{b}\cdot\mathbf{c}_2^\perp}{\|\mathbf{c}_2^\perp\|^2}\mathbf{c}_2^\perp.$$

## 2.5 Worked Examples

**Example 2.5.1** (Unique Least-Squares Solution)**.** Find $\hat{\mathbf{x}}$ for $\begin{pmatrix}1 & 2\\ 1 & 1\\ 1 & -1\end{pmatrix}\mathbf{x} = \begin{pmatrix}6\\4\\1\end{pmatrix}$.

**Normal equations:**

$$A^TA = \begin{pmatrix}3 & 2\\ 2 & 6\end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix}11\\15\end{pmatrix}.$$

$\det(A^TA) = 14$. Then:

$$\hat{\mathbf{x}} = \frac{1}{14}\begin{pmatrix}6 & -2\\ -2 & 3\end{pmatrix}\begin{pmatrix}11\\15\end{pmatrix} = \begin{pmatrix}36/14\\23/14\end{pmatrix} = \begin{pmatrix}18/7\\23/14\end{pmatrix}.$$

**Error:** $\|A\hat{\mathbf{x}} - \mathbf{b}\| \approx 0.267$.

**Example 2.5.2** (Closest Point on a Line)**.** Find the point on $y = 3x$ closest to $(2, 3)$.

Matrix form: $\begin{pmatrix}1\\3\end{pmatrix} t = \begin{pmatrix}2\\3\end{pmatrix}$.

Normal equation: $10\hat{t} = 11 \implies \hat{t} = \frac{11}{10} = 1.1$.

**Closest point:** $(1.1,\ 3.3)$.

**Example 2.5.3** (Closest Point on a Plane)**.** Find the point on the plane $2x + 3y = z$ closest to $\mathbf{b} = (1, 0, 0)$.

The plane has normal $(2, 3, -1)$, but as a subspace $\{(x, y, 2x + 3y)\}$, it is spanned by $\mathbf{c}_1 = (1, 0, 2)$ and $\mathbf{c}_2 = (0, 1, 3)$.

$A = \begin{pmatrix}1 & 0\\ 0 & 1\\ 2 & 3\end{pmatrix}$. Normal equations: $A^TA = \begin{pmatrix}5 & 6\\ 6 & 10\end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix}1\\0\end{pmatrix}$.

$\det(A^TA) = 14$. $\hat{\mathbf{x}} = \frac{1}{14}\begin{pmatrix}10 & -6\\ -6 & 5\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}5/7\\-3/7\end{pmatrix}$.

**Closest point:** $A\hat{\mathbf{x}} = \begin{pmatrix}5/7\\-3/7\\-1/7\end{pmatrix}$.

**Example 2.5.4** (Infinitely Many Solutions)**.** $\begin{pmatrix}1 & 2\\ 1 & 2\end{pmatrix}\mathbf{x} = \begin{pmatrix}4\\3\end{pmatrix}$. The columns are dependent, so $A^TA = \begin{pmatrix}2 & 4\\ 4 & 8\end{pmatrix}$ is singular. Solutions: $\hat{\mathbf{x}} = \begin{pmatrix}7/2 - 2t\\ t\end{pmatrix}$ for $t \in \mathbb{R}$. All give the same projection (and error).

## 2.6 Common Misconceptions

*Remark* 2.6.1 (Why $(A^TA)^{-1}A^T \neq A^{-1}$). Writing $(A^TA)^{-1}A^T = A^{-1}(A^T)^{-1}A^T = A^{-1}$ is **wrong** unless $A$ is square and invertible. In the least-squares setting, $A$ is typically $m \times n$ with $m > n$ (not square), so $A^{-1}$ and $(A^T)^{-1}$ do not exist.

*Remark* 2.6.2 (Switching Columns). Switching columns of $A$ (multiplying by a permutation $P$: $A' = AP$) does not change $\mathrm{Col}(A)$ or the minimum error. The solution changes as $\hat{\mathbf{x}}' = P^T\hat{\mathbf{x}}$ (just reorders components).

# Chapter 3

# Curve Fitting

## 3.1 Least-Squares Line

Given points $(x_1, y_1), \ldots, (x_n, y_n)$, fit $f(x) = mx + b$. Writing one equation per point:

$$\underbrace{\begin{pmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix}}_{A} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \mathbf{b}.$$

Solve the normal equation $A^T A\hat{\mathbf{c}} = A^T\mathbf{b}$. This minimizes $\sum(y_i - f(x_i))^2$ (sum of squared *vertical* distances).

*Remark* 3.1.1. The line $y = mx+b$ and the line $x = ny+c$ (which minimizes horizontal distances) are generally **different** because they minimize different objective functions.

## 3.2 General Function Fitting: What Works?

Least squares applies whenever the unknown coefficients appear **linearly**:

$$f(x) = a_1 f_1(x) + \cdots + a_k f_k(x),$$

where $f_1, \ldots, f_k$ are known functions. The matrix is:

$$A = \begin{pmatrix} f_1(x_1) & \cdots & f_k(x_1) \\ \vdots & & \vdots \\ f_1(x_n) & \cdots & f_k(x_n) \end{pmatrix}.$$

**Example 3.2.1** (What Works and What Doesn't)**.** Matrix column $i$: $(e^{x_i}, x_i)$.

- $f(x) = ae^x + bx$: **Yes** (linear in $a, b$).
- $f(x) = a\sin x + b\cos x + c$: **Yes** (linear in $a, b, c$).
- $f(x) = e^{ax} + bx$: **No** ($a$ appears in exponent).
- $f(x) = a\sin(bx) + c$: **No** ($b$ appears inside sine).
- $f(x) = ax^2 + bx^c$: **No** ($c$ appears as exponent).

## 3.3 Common Fitting Templates

### Polynomial Fit

**Parabola** $f(x) = ax^2 + bx + c$: column $i$ of $A$ is $(x_i^2, x_i, 1)$.
**Degree-$k$ polynomial**: column $i$ of $A$ is $(x_i^k, x_i^{k-1}, \ldots, x_i, 1)$.

### Exponential Fit

$f(x) = ae^x + b$: column $i$ of $A$ is $(e^{x_i}, 1)$.

### Sinusoidal Fit

$f(x) = a + b\sin x$: column $i$ of $A$ is $(1, \sin x_i)$.

### Surface Fitting

For $f(x, y) = a_1 g_1(x, y) + \cdots + a_k g_k(x, y)$, each data point $(x_i, y_i, z_i)$ gives one row. Example: paraboloid $f(x, y) = ax^2 + by^2$, matrix row $i$: $(x_i^2, y_i^2)$.

### Ellipse Fitting (Polar Form)

For an ellipse $Ar^2\cos^2\theta + Br^2\sin^2\theta = 1$, rewrite as:

$$(r_i\cos\theta_i)^2 \cdot A + (r_i\sin\theta_i)^2 \cdot B = 1.$$

Matrix row $i$: $(r_i^2\cos^2\theta_i,\ r_i^2\sin^2\theta_i)$, right-hand side all 1s.

## 3.4 Worked Examples

**Example 3.4.1** (Line Through Three Points)**.** Points $(-1, 0), (1, 2), (2, 2)$.

$$A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}.$$

$A^TA = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$, det $= 14$.

$$\hat{\mathbf{c}} = \frac{1}{14}\begin{pmatrix} 3 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 11/7 \end{pmatrix}.$$

**Line:** $f(x) = \frac{1}{2}x + \frac{11}{7} \approx 0.5x + 1.571$.

**Example 3.4.2** (Parabola Fit)**.** Points $(-1, 3), (0, 1), (1, 2), (3, 9)$. Set up $A$ with columns $(x_i^2, x_i, 1)$ and solve. The least-squares parabola is approximately $f(x) \approx 1.045x^2 - 0.555x + 1.291$.

**Example 3.4.3** (Sinusoidal Fit)**.** Points $(-2, 6.3), (3, 1.2), (5, 7.1), (8, -2.8), (9, -0.05)$. Fit $f(x) = a + b\sin x$.

$$A = \begin{pmatrix} 1 & \sin(-2) \\ 1 & \sin 3 \\ 1 & \sin 5 \\ 1 & \sin 8 \\ 1 & \sin 9 \end{pmatrix}.$$

Solving: $f(x) \approx 2.026 - 4.974\sin x$. Predict $f(\pi/2) \approx -2.948$.

## 3.5 Using the Best-Fit Function

After fitting:

1. Evaluate $f$ at any point to make predictions.
2. Solve $f(x) = c$ to find when the function reaches a target.
3. Compute $f'$ to find rates of change.
4. Compare the least-squares errors of competing models to select the best fit.

# Chapter 4

# Team Ranking: Massey's Method

## 4.1 Setup

Given $T$ teams and game outcomes (each described by a winning margin), assign rankings $r_1, \ldots, r_T$.

**Key premise:** If team $i$ beats team $j$ by $p$ points, the rankings should satisfy $r_i - r_j = p$.

For each game, write one equation. This gives $A\mathbf{r} = \mathbf{p}$ where each row of $A$ has $+1$ in column $i$, $-1$ in column $j$, and 0 elsewhere.

Since game results are often inconsistent, $A\mathbf{r} = \mathbf{p}$ rarely has an exact solution. We use least squares.

## 4.2 Massey's Method

### The Massey Matrix

The least-squares normal equation is $M\hat{\mathbf{r}} = \bar{q}$ where $M = A^T A$ and $\bar{q} = A^T \mathbf{p}$.

**Structure of $M$:**

- $m_{ii}$ = total games team $i$ played.
- $m_{ij} = -($ number of games between teams $i$ and $j)$ for $i \neq j$.
- $M$ is symmetric. Its rows all sum to zero $\Rightarrow$ $M$ is singular.

**Structure of $\bar{q}$:**

- $q_i$ = sum of all point margins for team $i$ (wins positive, losses negative).
- Note: $\sum_i q_i = 0$ always (each game contributes $+p$ to one team and $-p$ to the other).

**Fact 4.2.1.** *The sum of diagonal entries of $M$ is always even, since it equals $2G$ (twice the total number of games $G$), as each game contributes $+1$ to exactly two diagonal entries.*

### Fixing Singularity: The Sum Constraint

Since $M$ is singular, modify as follows:

1. Replace the **last row** of $M$ with all 1s.
2. Replace the **last entry** of $\bar{q}$ with 0.

This enforces $r_1 + \cdots + r_T = 0$ (rankings centered at zero). The modified system is invertible and has a unique solution.

## 4.3 Shortcut: Building $M\hat{\mathbf{r}} = \bar{q}$ Directly

Rather than assembling $A$ first:

- Set $m_{ii}$ = games played by team $i$ (for $i < T$).
- Set $m_{ij} = -($ games between $i$ and $j)$ for $i \neq j$ (for $i < T$).
- Last row: all 1s.
- Set $q_i$ = total points won by team $i$ (for $i < T$).
- Last entry of $\bar{q}$: 0.

## 4.4 Worked Examples

**Example 4.4.1** (Four Teams)**.** Games: T1–T2 (1 game), T1–T3 (1), T2–T3 (2), T2–T4 (1), T3–T4 (1). Results: T1 beats T2 by 7, T1 loses to T3 by 8, T2 ties T3, T2 beats T3 by 15, T2 beats T4 by 5, T3 loses to T4 by 1.

Degrees: T1 plays 2, T2 plays 4, T3 plays 4, T4 plays 2. Point totals: $q_1 = 7 - 8 = -1$, $q_2 = -7 + 0 + 15 + 5 = 13$, $q_3 = 8 + 0 - 15 - 1 = -8$, $q_4 = -5 + 1 = -4$.

$$M = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 4 & -2 & -1 \\ -1 & -2 & 4 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \bar{q} = \begin{pmatrix} -1 \\ 13 \\ -8 \\ 0 \end{pmatrix}.$$

Solving: $\hat{r}_1 \approx 0.125$, $\hat{r}_2 \approx 2.375$, $\hat{r}_3 \approx -1.125$, $\hat{r}_4 \approx -1.375$.

Order: **T2** > **T1** > **T3** > **T4**.

Prediction T1 vs T4 (never played): $r_1 - r_4 = 1.5$ (T1 expected to win by 1.5).

## 4.5 Special Situations

### Symmetric Round Robin

If T1 beats T2 by $k$, T2 beats T3 by $k$, T3 beats T1 by $k$: the point totals are all zero ($q_1 = k - k = 0$, etc.), giving all equal rankings $r_1 = r_2 = r_3 = 0$. The cyclic symmetry of the results implies no team has a net advantage.

### Effect of Repetition

If T1 beats T2 by $n$ (once) vs. beats T2 by 1 ($n$ times):

- **Once by $n$:** $r_1 = (n-1)/3$ (grows without bound as $n \to \infty$).
- **$n$ times by 1:** $r_1 = (n-1)/(2n+1) \to 1/2$ (bounded).

One large win counts more than many small wins.

### Home Advantage

Reduce winning margins for home victories by a fixed percentage (e.g., 25%) before entering the margins into the Massey system.

### Disconnected Team Graphs

If the teams split into two groups with no games between groups, the Massey matrix (before modification) is block-singular and cannot be solved globally. Apply Massey separately to each connected group.

## 4.6 Prediction Formula

If teams $i$ and $j$ have never played, the expected margin when they meet is:

$$\hat{p}_{ij} = r_i - r_j.$$

A positive value means team $i$ is expected to win.

# Chapter 5

# Graph Theory and the Fiedler Method

## 5.1 Graph Basics

**Definition 5.1.1** (Graph, Simple Graph, Connected)**.** A **graph** $G = (V, E)$ is a set of **vertices** (nodes) $V$ and **edges** $E$ (pairs of vertices). A graph is:

- **Simple**: no loops (edges from a vertex to itself) and no multiple edges.
- **Connected**: there is a path between every pair of vertices.
- $k$**-regular**: every vertex has exactly $k$ neighbors (degree $k$).

## 5.2 Matrix Representations

**Definition 5.2.1** (Adjacency Matrix)**.** The **adjacency matrix** $A$ of a simple graph on $n$ vertices is the $n \times n$ symmetric matrix with $a_{ij} = 1$ if $\{i, j\} \in E$ and $a_{ij} = 0$ otherwise.

**Definition 5.2.2** (Degree Matrix and Laplacian)**.**

$$D = \operatorname{diag}(\deg v_1, \ldots, \deg v_n), \qquad L = D - A.$$

$L$ is the **Laplacian matrix**.

**Example 5.2.1.** For the path graph $1 - 2 - 3$:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

## 5.3 Walks and Powers of $A$

**Definition 5.3.1** (Walk)**.** A **walk** of length $k$ from vertex $i$ to vertex $j$ is a sequence of $k$ edges starting at $i$ and ending at $j$ (vertices and edges may repeat).

**Theorem 5.3.1** (Walks via $A^k$)**.** $(A^k)_{ij}$ = *number of walks of length $k$ from vertex $i$ to vertex $j$.*

*Proof.* Induction on $k$. Base: $(A^1)_{ij} = a_{ij}$ = number of walks of length 1 (0 or 1). Step: $(A^{k+1})_{ij} = \sum_\ell (A^k)_{i\ell} a_{\ell j}$ sums walks of length $k$ to $\ell$ followed by one edge $\ell \to j$. □

**Definition 5.3.2** (Trace)**.** $\operatorname{tr}(M) = \sum_i m_{ii}$.

**Theorem 5.3.2** (Triangle Count)**.** *Number of triangles* $= \frac{1}{6}\operatorname{tr}(A^3)$.

*Proof.* A closed walk of length 3 from $v$ to $v$ in a simple graph must traverse exactly one triangle. Each triangle has 3 vertices and can be traversed in 2 directions, contributing 2 to each of the 3 diagonal entries. So $\operatorname{tr}(A^3) = 6 \times (\text{triangles})$. □

*Remark* 5.3.1 (Why Not $\operatorname{tr}(A^4)$ for Squares?). Closed walks of length 4 include both squares *and* backtracks ($v \to u \to v \to u \to v$) and triangle-backtracks. Unlike triangles, not all closed 4-walks are squares, so $\operatorname{tr}(A^4)$ overcounts. The triangle formula works because the only closed walk of length 3 in a simple graph *is* a triangle.

## 5.4 Graph Isomorphism

**Definition 5.4.1** (Isomorphism)**.** Graphs $G_1$ and $G_2$ (both on $n$ vertices) are **isomorphic** ($G_1 \cong G_2$) if there exists a permutation matrix $P$ such that $PA_1P^T = A_2$, i.e., the vertices can be relabeled to make the adjacency matrices equal.

**Theorem 5.4.1** (Triangle Count Invariant)**.** *If* $G_1 \cong G_2$ *then* $\frac{1}{6}\operatorname{tr}(A_1^3) = \frac{1}{6}\operatorname{tr}(A_2^3)$.

**Example 5.4.1** (Non-Isomorphic Graphs)**.** The triangle $K_3$ has adjacency matrix $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ (1 triangle), while the path $P_3$ has $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ (0 triangles). They are not isomorphic.

*Remark* 5.4.1. The converse does not hold: two non-isomorphic graphs can have the same triangle count. For example, $K_4$ and two disjoint $K_3$ triangles both have 4 triangles but are not isomorphic ($K_4$ is connected, $K_3 \sqcup K_3$ is not).

## 5.5 Graph Partitioning

**Definition 5.5.1** (Partition and Cut)**.** A **2-partition** $(V_1, V_2)$ of $V$ divides the vertices into two nonempty disjoint sets. The **cut** is the number of edges between $V_1$ and $V_2$.

**Goal:** Find a balanced partition (roughly $|V_1| \approx |V_2|$) with minimum cut.

**Lemma 5.5.1.** *Assign* $x_i = +1$ *for* $i \in V_1$ *and* $x_i = -1$ *for* $i \in V_2$. *Then* $\operatorname{cut}(V_1, V_2) = \frac{1}{4}\sum_{\{i,j\}\in E}(x_i - x_j)^2$.

## 5.6 The Fiedler Method

**Fact 5.6.1** (Laplacian Eigenvalues)**.** *For a simple connected graph on $n$ vertices, the eigenvalues of $L$ satisfy:*

$$0 = \lambda_1 < \lambda_2 \le \lambda_3 \le \cdots \le \lambda_n.$$

*The smallest eigenvalue is always 0 (eigenvector* $\mathbf{1} = (1, 1, \ldots, 1)^T$*). The graph is connected* $\iff \lambda_2 > 0$.

**Definition 5.6.1** (Fiedler Value and Fiedler Vector)**.**

- The **Fiedler value** (algebraic connectivity) is $\lambda_2$.
- A **Fiedler vector** is any eigenvector corresponding to $\lambda_2$.

**Theorem 5.6.1** (Fiedler Partitioning Method)**.** *Compute the Fiedler vector* $\mathbf{v}$*. Partition:*

$$V_+ = \{i : v_i > 0\}, \qquad V_- = \{i : v_i < 0\}.$$

*Vertices with* $v_i = 0$ *may go in either group.*

**Algorithm:**

1. Build $L = D - A$.
2. Compute eigenpairs of $L$; sort eigenvalues.
3. The eigenvector for the second smallest eigenvalue is the Fiedler vector.
4. Partition by sign.

**Example 5.6.1** (Triangle with Pendant)**.** Graph: 1–2, 1–3, 2–3, 3–4.

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

Eigenvalues: $0, 1, 3, 4$. Fiedler vector: $\approx (-0.408, -0.408, 0, 0.816)^T$.

Partition: $V_- = \{1, 2\}$ (negative), $V_0 = \{3\}$ (zero, bridge vertex), $V_+ = \{4\}$ (positive). This correctly identifies the triangle group vs. the pendant vertex.

**Example 5.6.2** (Complete Graph $K_n$)**.** The Laplacian of $K_n$ is $L = nI - J$ where $J$ is the all-ones matrix. Eigenvalues: 0 (once) and $n$ ($n - 1$ times). So $\lambda_2 = n$ and the Fiedler value equals the number of vertices.

*Remark* 5.6.1 (Useful Facts about the Fiedler Vector).

- Since $L$ is symmetric, it has real eigenvalues and an orthonormal eigenbasis.
- The Fiedler vector is orthogonal to $\mathbf{1}$, so its entries sum to 0.
- The partition produced approximately minimizes cut while balancing group sizes.
- If the eigenspace for $\lambda_2$ has dimension $> 1$, multiple Fiedler vectors exist; each can give a (potentially different but equally valid) partition.

# Chapter 6

# Markov Chains

This chapter covers Markov chains from two sources: (1) Justin Wyss-Gallifent's notes (Ch. 6), and (2) Lay & McDonald, *Linear Algebra and Its Applications*, Chapter 10, Sections 10.1–10.3.

## 6.1 Introduction and Motivation

### Two-State Example

Suppose a total population of 1 is split between two states: 0.7 in State 1 and 0.3 in State 2. Each year, 10% of State 1 moves to State 2, and 5% of State 2 moves to State 1. After one year:

$$\text{State 1} = 0.90(0.7) + 0.05(0.3) = 0.645, \qquad \text{State 2} = 0.10(0.7) + 0.95(0.3) = 0.355.$$

In matrix form with transition matrix $T$:

$$T = \begin{pmatrix} 0.90 & 0.05 \\ 0.10 & 0.95 \end{pmatrix}, \quad \mathbf{x}_1 = T\mathbf{x}_0, \quad \mathbf{x}_n = T^n\mathbf{x}_0.$$

**Column convention:** Column $j$ of $T$ is an "exit vector" from state $j$. Entry $T_{ij}$ = fraction moving from state $j$ to state $i$ in one step.

### Lay-McDonald 10.1: Finite-State Markov Chains

A **finite-state Markov chain** is a mathematical model satisfying:

1. The next state depends **only** on the current state (Markov property – no memory).
2. Transition probabilities do not change with time (time-homogeneous).

Applications include signal transmission, gas diffusion (Ehrenfest model), and random walks.

## 6.2 Ehrenfest Diffusion Model (Lay-McDonald 10.1)

**Setup:** Two urns (A and B) containing $2k$ molecules total. At each step, pick one molecule uniformly at random and move it to the other urn. Track the number of molecules in urn A.

For $k = 3$ (6 molecules), states are $0, 1, \ldots, 6$. Transition probabilities from state $i$:

$$P(i \to i+1) = \frac{6-i}{6}, \qquad P(i \to i-1) = \frac{i}{6}.$$

States 0 and 6 transition deterministically to 1 and 5 respectively. The resulting $7 \times 7$ transition matrix has a checkered structure.

*Remark* 6.2.1. The Ehrenfest model does not reach a unique steady state because it is not regular (the transition matrix alternates between two classes of states). Long-term behavior depends on the parity of the initial state.

## 6.3 Random Walks (Lay-McDonald 10.1)

**Random walk on $\{1, 2, \ldots, n\}$:** At each step, move left with probability $p$ or right with probability $1 - p$.

- **Absorbing boundaries:** States 1 and $n$ are absorbing (once reached, stay there).
- **Reflecting boundaries:** States 1 and $n$ bounce back inward.

Transition matrices:

$$P_{\text{absorbing}} = \begin{pmatrix} 1 & p & 0 & \cdots \\ 0 & 0 & p & \cdots \\ 0 & 1-p & 0 & \cdots \\ \vdots & & & \ddots \end{pmatrix}, \qquad P_{\text{reflecting}} = \begin{pmatrix} 0 & p & 0 & \cdots \\ 1 & 0 & p & \cdots \\ 0 & 1-p & 0 & \cdots \\ \vdots & & & \ddots \end{pmatrix}.$$

**Random walk on a directed graph:** States are vertices; at each step, follow a uniformly random outgoing edge. This is the basis for Google PageRank.

## 6.4 Steady States and Limits

**Definition 6.4.1** (Probability Vector)**.** A vector $\mathbf{x}$ with non-negative entries summing to 1.

**Definition 6.4.2** (Transition Matrix (Stochastic Matrix))**.** A matrix $T$ whose columns are each probability vectors. Entry $T_{ij}$ = fraction of population moving from state $j$ to state $i$ per step.

**Definition 6.4.3** (Steady-State Vector)**.** A probability vector $\mathbf{x}^*$ with $T\mathbf{x}^* = \mathbf{x}^*$, i.e., an eigenvector of $T$ for eigenvalue $\lambda = 1$.

**Definition 6.4.4** (Regular Transition Matrix)**.** $T$ is **regular** if $T^k$ has all positive entries for some $k \geq 1$.

**Theorem 6.4.1** (Convergence – Wyss-Gallifent & Lay-McDonald 10.2)**.** *If $T$ is a regular transition matrix, then:*

1. *$T$ has a **unique** steady-state vector $\mathbf{x}^*$.*
2. *For any probability vector $\mathbf{x}_0$: $\lim_{k\to\infty} T^k \mathbf{x}_0 = \mathbf{x}^*$.*
3. *$\lim_{k\to\infty} T^k$ exists and has all columns equal to $\mathbf{x}^*$.*

*Proof of (3) given (2).* Let $\mathbf{e}_i$ be the $i$-th standard basis vector. By (2), $\lim_{k\to\infty} T^k \mathbf{e}_i = \mathbf{x}^*$ for each $i$. Hence:

$$\lim_{k\to\infty} T^k = \lim_{k\to\infty} T^k [\mathbf{e}_1 \ \cdots \ \mathbf{e}_n] = [\mathbf{x}^* \ \cdots \ \mathbf{x}^*]. \quad \square$$

$\square$

*Remark* 6.4.1 (Non-Regular Matrices)*.* A non-regular $T$ may have multiple steady-state vectors or no convergence. Examples:

- $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: oscillates forever, does not converge except from $\mathbf{x}_0 = (0.5, 0.5)^T$.
- A block upper-triangular $T$: population drains into absorbing bottom-right block.

### Meaning of $T^k_{ij}$

The $(i, j)$-entry of $T^k$ equals the fraction of the population starting in state $j$ that is in state $i$ after exactly $k$ steps. This follows from the matrix product formula for $T^k$.

## 6.5 Finding the Steady State

**Method 1 (Eigenvalue):** Solve $(T - I)\mathbf{x} = \mathbf{0}$ with $\sum x_i = 1$. Software gives a unit eigenvector; divide by its entry-sum to get the probability vector.

**Method 2 (Iteration):** Compute $T^k\mathbf{x}_0$ for large $k$; the result converges to $\mathbf{x}^*$ by Theorem 6.4.1.

**Method 3 (Column of $T^k$):** Pick any column of $T^k$ for large $k$; it converges to $\mathbf{x}^*$.

**Example 6.5.1** (Three States)**.** $T = \begin{pmatrix} 0.90 & 0.10 & 0.20 \\ 0.06 & 0.80 & 0.10 \\ 0.04 & 0.10 & 0.70 \end{pmatrix}$. Eigenvalues: $1, 0.763, 0.637$. Eigenvector for $\lambda = 1$ (normalized): $\mathbf{x}^* \approx (0.581, 0.256, 0.163)^T$.

## 6.6 Proof for the $2 \times 2$ Case

**Theorem 6.6.1.** *If* $T = \begin{pmatrix} a & 1-b \\ 1-a & b \end{pmatrix}$ *is a regular transition matrix* ($0 < a, b < 1$, $(a, b) \neq (0, 0), (1, 1)$)*, then* $\lim_{n\to\infty} T^n\mathbf{x}_0 = \mathbf{x}^*$ *for all probability vectors* $\mathbf{x}_0$*.*

*Proof.* The eigenpairs of $T$ are:

$$\left(\lambda_1 = 1,\ \mathbf{v}_1 = \begin{pmatrix} b-1 \\ a-1 \end{pmatrix}\right), \qquad \left(\lambda_2 = a + b - 1,\ \mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}\right).$$

Regularity forces $-1 < \lambda_2 = a + b - 1 < 1$, so $\lambda_2^n \to 0$. Diagonalizing:

$$T^n = P \begin{pmatrix} 1 & 0 \\ 0 & \lambda_2^n \end{pmatrix} P^{-1} \xrightarrow{n\to\infty} P \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} P^{-1} = \begin{pmatrix} \frac{b-1}{a+b-2} & \frac{b-1}{a+b-2} \\ \frac{a-1}{a+b-2} & \frac{a-1}{a+b-2} \end{pmatrix}.$$

Multiplying by any probability vector $\mathbf{x}_0 = \begin{pmatrix} c \\ 1-c \end{pmatrix}$ yields $\begin{pmatrix} \frac{b-1}{a+b-2} \\ \frac{a-1}{a+b-2} \end{pmatrix} = \mathbf{x}^*$ (independent of $c$).

□ □

## 6.7 Communication Classes (Lay-McDonald 10.3)

**Definition 6.7.1** (Communicating States)**.** States $i$ and $j$ **communicate** (written $i \leftrightarrow j$) if the chain can reach state $i$ from state $j$ *and* state $j$ from state $i$ (in some number of steps). Communication is an equivalence relation, and its equivalence classes are called **communication classes**.

**Definition 6.7.2** (Absorbing State)**.** State $i$ is **absorbing** if $T_{ii} = 1$ (once entered, never left). Equivalently, state $i$ forms its own communication class and has no outgoing transitions to other states.

**Definition 6.7.3** (Absorbing Markov Chain)**.** A Markov chain is an **absorbing Markov chain** if:

1. It has at least one absorbing state, and
2. From every non-absorbing state, it is possible to reach some absorbing state.

**Theorem 6.7.1** (Absorbing Chains Converge)**.** *In an absorbing Markov chain, the chain is absorbed with probability 1 (eventually reaches an absorbing state and stays there).*

**Example 6.7.1** (Random Walk with Absorbing Boundaries)**.** For the random walk on $\{1, 2, 3, 4, 5\}$ with states 1 and 5 absorbing: states 2, 3, 4 are transient (eventually leave and never return). The chain is always eventually absorbed at state 1 or state 5. The probability of absorption at state 1 vs. state 5 depends on the starting state and the step probability $p$.

*Remark* 6.7.1 (Standard Form for Absorbing Chains). Reorder states so absorbing states come first. The transition matrix takes the standard form:

$$P = \begin{pmatrix} I & R \\ 0 & Q \end{pmatrix}$$

where $I$ is the identity on absorbing states, $Q$ is the transition matrix among transient states, and $R$ is the matrix of transitions from transient to absorbing states.

*Remark* 6.7.2 (Non-Absorbing Non-Regular Chains). A chain can be neither regular nor absorbing. The Ehrenfest diffusion model is an example: it does not converge to a steady state from all starting points, yet it has no absorbing states. Such chains typically have periodic behavior.

## 6.8 Directed Graphs and Markov Chains (Lay-McDonald 10.1)

**Definition 6.8.1** (Directed Graph)**.** A **directed graph** (digraph) has edges with direction: an edge from $i$ to $j$ means we can go from $i$ to $j$ but not necessarily from $j$ to $i$.

The transition matrix for a random walk on a directed graph: if vertex $j$ has out-degree $d_j$, then $T_{ij} = 1/d_j$ if there is an edge $j \to i$, and $T_{ij} = 0$ otherwise.

**Example 6.8.1** (Signal Transmission)**.** Each bit (0 or 1) passes through a noisy channel. With probability $p$, a bit passes correctly; with probability $1 - p$, it flips. The transition matrix is:

$$P = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}.$$

After two stages, the probability a 0 remains a 0 is $(P^2)_{00} = p^2 + (1-p)^2$. For $p = 0.99$: $(P^2)_{00} = 0.9802$.

# Chapter 7

# Google PageRank

## 7.1 The Random Websurfer Model

PageRank models a **Random Websurfer (RW)** navigating the internet:

1. The RW is at some page.
2. If the page has outbound links: with probability 0.85 the RW follows a uniformly random link; with probability 0.15 the RW jumps to a uniformly random page anywhere.
3. If the page has **no outbound links** (dangling node): the RW jumps uniformly to any page.

This is a Markov chain. The state space is the set of all webpages.

## 7.2 The PageRank Transition Matrix (Lay-McDonald 10.2)

**Definition 7.2.1** (Link Matrix $S$)**.** Let $n$ be the number of pages. The **link matrix** $S$ is $n \times n$ with:

$$S_{ij} = \begin{cases} 1/k & \text{page } j \text{ links to page } i \text{ and } j \text{ has } k \text{ outbound links,} \\ 1/n & \text{page } j \text{ has no outbound links (dangling node).} \end{cases}$$

**Definition 7.2.2** (Google Matrix / PageRank Transition Matrix)**.**

$$T = \frac{0.15}{n}\mathbf{1}\mathbf{1}^T + 0.85\,S,$$

where $\mathbf{1}\mathbf{1}^T$ is the $n \times n$ all-ones matrix $J$.

**Theorem 7.2.1.** *$T$ is a regular transition matrix (all entries $\geq 0.15/n > 0$, so $T^1$ has all positive entries).*

By Theorem 6.4.1 (the Convergence Theorem for regular Markov chains), $T$ has a unique steady-state vector $\mathbf{x}^*$, which is the **PageRank vector**.

**Definition 7.2.3** (PageRank)**.** The **PageRank** of page $i$ is the $i$-th entry of the steady-state vector $\mathbf{x}^*$ satisfying $T\mathbf{x}^* = \mathbf{x}^*$, $\sum x_i^* = 1$.

## 7.3 Interpretation

The PageRank of page $i$ equals the long-run fraction of time the Random Websurfer spends at page $i$. Pages with higher PageRank are considered more important.

**Why this is better than counting inbound links:**

- Being linked by an important page contributes more than being linked by a trivial one.
- Importance is defined *recursively*: a page is important if important pages link to it.
- This self-referential definition is captured by $T\mathbf{x}^* = \mathbf{x}^*$ (eigenvector equation).

*Remark* 7.3.1 (Outbound Links). Outbound links affect where traffic *leaves* your page, not where it arrives. Adding or removing outbound links does not change your PageRank.

## 7.4 Computing PageRank

**Eigenvector method:** Solve $T\mathbf{x} = \mathbf{x}$ (i.e., $(T - I)\mathbf{x} = \mathbf{0}$) with the constraint $\sum x_i = 1$. Since $T$ is regular, $\lambda = 1$ is the unique eigenvalue of maximum modulus.

**Power iteration:** Start with any probability vector $\mathbf{x}_0$ (e.g., $\mathbf{x}_0 = (1/n, \ldots, 1/n)^T$) and compute $\mathbf{x}_{k+1} = T\mathbf{x}_k$. Since $T$ is regular, this converges to $\mathbf{x}^*$.

**Scalability note:** For the actual web ($n \sim 10^{10}$), we do not compute eigenvalues of $T$ (an $n \times n$ characteristic polynomial). Power iteration only requires matrix-vector products, which is far more tractable.

## 7.5 Special Cases and Examples

**Example 7.5.1** (Two Symmetric Pages)**.** P1 $\leftrightarrow$ P2. By symmetry, both pages have PageRank 0.5.

**Example 7.5.2** (Fully Connected Internet)**.** All $n$ pages link to all $n - 1$ others. The uniform vector $\mathbf{u} = (1/n, \ldots, 1/n)^T$ satisfies $T\mathbf{u} = \mathbf{u}$. All pages have equal PageRank $1/n$.

**Example 7.5.3** (Dangling Node)**.** Page $n$ has no outbound links. Its column in $S$ is $(1/n, \ldots, 1/n)^T$ (uniform random jump). The 0.15 random-jump term already covers 15% of this; the remaining 85% becomes the dangling contribution.

**Example 7.5.4** (Disconnected Internet)**.** Without the random-jump term (i.e., $d = 1$), a disconnected internet may have no regular transition matrix and no unique steady state. The random jump ($d = 0.85$) adds $0.15/n > 0$ to every entry of $T$, making it regular and ensuring convergence.

## 7.6 Effect of the Damping Factor

The damping factor $d = 0.85$ (probability of following a link vs. jumping randomly) was chosen by Brin and Page empirically. Changing $d$ generally preserves the ranking order but affects the magnitudes of PageRanks. As $d \to 1$, the ranking becomes more sensitive to link structure; as $d \to 0$, it approaches uniform $1/n$.

## 7.7 Chain Internet

For the chain $P_1 \to P_2 \to \cdots \to P_n$: page $P_n$ is a dangling node (no outbound links), contributing random jumps to all pages. The transition matrix is:

$$T = \frac{0.15}{n} J + 0.85\, S, \quad S_{ij} = \begin{cases} 1 & j \to i \text{ is an edge, } j < n \\ 1/n & j = n \end{cases}.$$

# Chapter 8

# Spectral Theorem

*Source: Lay & McDonald, Linear Algebra and Its Applications, §7.1.*

*Remark* 8.0.1 (Suggested Problems 4). **Quiz:** Lay §7.1.13–22 (orthogonal diagonalization). **Exam:** Lay §7.1.25–38 (true/false and proofs).

## 8.1 Symmetric Matrices and Orthogonal Diagonalization

**Definition 8.1.1** (Symmetric Matrix)**.** A matrix $A$ is **symmetric** if $A^T = A$. Symmetric matrices are square.

**Theorem 8.1.1** (Eigenvectors of Symmetric Matrices)**.** *If $A$ is symmetric, then any two eigenvectors from different eigenspaces are orthogonal.*

**Definition 8.1.2** (Orthogonally Diagonalizable)**.** An $n \times n$ matrix $A$ is **orthogonally diagonalizable** if there exist an orthogonal matrix $P$ and a diagonal matrix $D$ such that $A = PDP^T = PDP^{-1}$.

**Theorem 8.1.2** (Spectral Theorem)**.** *An $n \times n$ matrix $A$ is orthogonally diagonalizable* if and only if *$A$ is symmetric. Equivalently: $A$ has $n$ real eigenvalues (counting multiplicities); the dimension of each eigenspace equals the multiplicity of the corresponding eigenvalue; eigenspaces for different eigenvalues are mutually orthogonal.*

## 8.2 Spectral Decomposition

If $A = PDP^T$ with orthonormal eigenvectors $u_1, \ldots, u_n$ and eigenvalues $\lambda_1, \ldots, \lambda_n$, then

$$A = \lambda_1 u_1 u_1^T + \lambda_2 u_2 u_2^T + \cdots + \lambda_n u_n u_n^T.$$

Each term $\lambda_i u_i u_i^T$ is a rank-1 matrix; $u_i u_i^T$ is the orthogonal projection onto $\operatorname{span}\{u_i\}$.

## 8.3 Repeated Eigenvalues

When an eigenvalue has multiplicity $> 1$, the eigenspace has dimension equal to the multiplicity. Use Gram–Schmidt on a basis of the eigenspace to obtain orthonormal eigenvectors for the orthogonal diagonalization.

## 8.4 How to Orthogonally Diagonalize a Symmetric Matrix

For a symmetric matrix $A$, the workflow is always the same:

1. Compute the eigenvalues of $A$.
2. For each eigenvalue $\lambda$, find a basis of $\mathrm{Nul}(A - \lambda I)$.
3. If an eigenspace has dimension greater than 1, apply Gram–Schmidt inside that eigenspace to obtain an orthonormal basis.
4. Normalize all eigenvectors and place them as columns of $P$.
5. Put the corresponding eigenvalues on the diagonal of $D$.

Then $A = PDP^T$ with $P^TP = I$.

*Remark* 8.4.1. The two checks that matter most in problems are:

$$P^TP = I \qquad \text{and} \qquad A = PDP^T.$$

If both hold, the orthogonal diagonalization is correct.

## 8.5 Worked Example

Orthogonally diagonalize

$$A = \begin{pmatrix} 7 & 2 \\ 2 & 4 \end{pmatrix}.$$

Since $A^T = A$, the Spectral Theorem applies.

**Step 1: Find the eigenvalues.**

$$\det(A - \lambda I) = \begin{vmatrix} 7-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix} = (7-\lambda)(4-\lambda) - 4 = \lambda^2 - 11\lambda + 24.$$

So

$$\lambda^2 - 11\lambda + 24 = (\lambda - 8)(\lambda - 3),$$

hence the eigenvalues are $\lambda_1 = 8$ and $\lambda_2 = 3$.

**Step 2: Find orthonormal eigenvectors.**

For $\lambda_1 = 8$,

$$A - 8I = \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix},$$

so $-x + 2y = 0$, hence $x = 2y$. One eigenvector is

$$v_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Normalize:

$$u_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 3$,

$$A - 3I = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix},$$

so $2x + y = 0$, hence $y = -2x$. One eigenvector is

$$v_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Normalize:

$$u_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

**Step 3: Build $P$ and $D$.**

$$P = \begin{pmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & -2/\sqrt{5} \end{pmatrix}, \qquad D = \begin{pmatrix} 8 & 0 \\ 0 & 3 \end{pmatrix}.$$

Therefore

$$\boxed{A = PDP^T.}$$

## 8.6 Why This Matters for SVD

The SVD of a matrix $A$ is built from the symmetric matrices $A^TA$ and $AA^T$. The Spectral Theorem guarantees that these matrices have orthonormal eigenbases, which is exactly what makes the singular value decomposition possible.

# Chapter 9

# Singular Value Decomposition

*Sources: Justin Wyss-Gallifent, Singular Value Decomposition; Lay & McDonald, Linear Algebra and Its Applications, §7.4.* The Spectral Theorem (Ch. 7.1) is used to construct the SVD.

*Remark* 9.0.1 (Suggested Problems 4). **Quiz:** Lay §7.4.5–12; Justin 8.1. **Exam:** Lay §7.4.13–18.

## 9.1 Introduction

Factoring a matrix means writing it as a product of other matrices. For example, if $A$ is diagonalizable we can write $A = PDP^{-1}$, where $D$ contains eigenvalues and $P$ contains eigenvectors.

The **singular value decomposition (SVD)** is another factorization used in data compression, matrix approximation, pseudoinverses, signal analysis, and recognition.

## 9.2 Definitions

**Definition 9.2.1** (Singular Value Decomposition)**.** For any $m \times n$ real matrix $A$, the **singular value decomposition** of $A$ is a factorization

$$A = U\Sigma V^T$$

where:

- $U$ is an $m \times m$ orthogonal matrix.
- $\Sigma$ is an $m \times n$ rectangular diagonal matrix with diagonal entries $\sigma_1, \ldots, \sigma_{\min(m,n)}$ all nonnegative and in nonincreasing order.
- $V$ is an $n \times n$ orthogonal matrix.

**Definition 9.2.2** (Orthogonal Matrix)**.** An $n \times n$ matrix $U$ is **orthogonal** if its columns are unit vectors all perpendicular to one another. Equivalently: columns form an orthonormal basis of $\mathbb{R}^n$, and $U^TU = UU^T = I$.

**Definition 9.2.3** (Rectangular Diagonal Matrix)**.** An $m \times n$ **rectangular diagonal** matrix has nonzero entries only in positions $(1, 1)$, $(2, 2)$, etc. Extra rows or columns (if $m \neq n$) are all zeros.

## 9.3 Constructing the SVD

### 9.3.1 Key Observations

**Observation 1.** If $k \leq \min(m, n)$ singular values are nonzero, then $A = U\Sigma V^T$ depends only on the first $k$ columns of $U$ and $V$. Moreover:

$$A\bar{v}_i = \sigma_i \bar{u}_i, \qquad A^T\bar{u}_i = \sigma_i \bar{v}_i \quad \text{for } 1 \leq i \leq k.$$

**Observation 2.** Consider $AA^T$ (size $m \times m$) and $A^TA$ (size $n \times n$):

- Both are symmetric; by the Spectral Theorem they have orthogonal eigenvectors.
- All eigenvalues are nonnegative: $\|A\bar{v}\|^2 = \bar{v}^T A^T A\bar{v} = \lambda\|\bar{v}\|^2 = \lambda$.
- There is a bijection between eigenpairs of $AA^T$ and $A^TA$ for nonzero eigenvalues:

$$(\sigma^2, \bar{u}) \mapsto \left(\sigma^2, \tfrac{1}{\sigma}A^T\bar{u}\right), \qquad (\sigma^2, \bar{v}) \mapsto \left(\sigma^2, \tfrac{1}{\sigma}A\bar{v}\right).$$

**Observation 3.** For each positive eigenvalue $\sigma_i^2$ common to both, we have paired unit eigenvectors $\bar{u}_i$ (left singular vector) and $\bar{v}_i$ (right singular vector) satisfying $A\bar{v}_i = \sigma_i\bar{u}_i$ and $A^T\bar{u}_i = \sigma_i\bar{v}_i$.

### 9.3.2 Construction in Brief

1. Find the positive eigenvalues of $AA^T$ (or $A^TA$). Denote them $\sigma_1^2 \geq \sigma_2^2 \geq \cdots \geq \sigma_k^2$.
2. For each $\sigma_i^2$, find unit eigenvector $\bar{u}_i$ of $AA^T$ and set $\bar{v}_i = \frac{1}{\sigma_i}A^T\bar{u}_i$ (or vice versa). Handle multiplicity by spanning eigenspaces.
3. Build $\Sigma$ with $\sigma_i$ on the diagonal; build $U$ from $\bar{u}_i$ and $V$ from $\bar{v}_i$. Fill remaining columns of $U$ and $V$ with eigenvectors for eigenvalue 0 to complete orthonormal bases.

*Remark* 9.3.1 (Which matrix should you use?). If $A$ is $m \times n$, then $A^TA$ is $n \times n$ and $AA^T$ is $m \times m$.

- Use the smaller one first when doing calculations by hand.
- The eigenvectors of $A^TA$ give the **right** singular vectors ($V$).
- The eigenvectors of $AA^T$ give the **left** singular vectors ($U$).

## 9.4 What to Show on an SVD Problem

When the problem asks you to *find an SVD*, the expected structure is:

1. Compute $A^TA$ or $AA^T$.
2. Find its eigenvalues and convert them into singular values $\sigma_i = \sqrt{\lambda_i}$.
3. Find orthonormal eigenvectors and assemble $V$ (or $U$).
4. Use $u_i = \frac{1}{\sigma_i}Av_i$ or $v_i = \frac{1}{\sigma_i}A^Tu_i$.
5. Write the final factorization $A = U\Sigma V^T$ and, if requested, verify by multiplication.

*Remark* 9.4.1. If a singular value is 0, the formula $u_i = \frac{1}{\sigma_i}Av_i$ does not apply. In that case, complete the relevant orthonormal basis using eigenvectors corresponding to the eigenvalue 0.

## 9.5 Worked Example

Find an SVD of

$$A = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix}.$$

**Step 1: Compute $A^T A$.**

$$A^T A = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}.$$

So the eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 0$, hence the singular values are

$$\sigma_1 = 2, \qquad \sigma_2 = 0.$$

**Step 2: Find $V$.** An orthonormal eigenbasis for $A^T A$ is

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

so we may take

$$V = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

**Step 3: Find the nonzero left singular vector.**

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{2} \begin{pmatrix} -2 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

To complete an orthonormal basis of $\mathbb{R}^2$, choose

$$u_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Thus

$$U = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

**Step 4: Build $\Sigma$.**

$$\Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

Therefore one SVD is

$$\boxed{A = U \Sigma V^T = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.}$$

## 9.6 Best Low-Rank Approximation

**Theorem 9.6.1** (Eckart–Young)**.** *Among all matrices of rank $\leq k$, the best approximation to $A$ (in Frobenius norm) that preserves the most variance is*

$$A_k = \sum_{i=1}^{k} \sigma_i \bar{u}_i \bar{v}_i^T = U \Sigma_k V^T,$$

*where $\Sigma_k$ is $\Sigma$ with $\sigma_{k+1}, \ldots, \sigma_{\min(m,n)}$ set to zero.*

The proportion of variance preserved is

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_{\min(m,n)}^2}.$$

*Remark* 9.6.1. This same quantity is what appears in image-compression problems as the **quality** or **variance preserved**. Keeping the first $k$ singular values gives the best rank-$k$ approximation in this sense.

## 9.7 Fundamental Subspaces and Reduced SVD (Lay-McDonald §7.4)

Given $A = U\Sigma V^T$ with rank $r$: $\{u_1, \ldots, u_r\}$ is an orthonormal basis for $\operatorname{Col} A$; $\{u_{r+1}, \ldots, u_m\}$ for $\operatorname{Nul} A^T$; $\{v_1, \ldots, v_r\}$ for $\operatorname{Row} A$; $\{v_{r+1}, \ldots, v_n\}$ for $\operatorname{Nul} A$.

**Reduced SVD:** $A = U_r D V_r^T$ where $U_r$, $V_r$ contain only the first $r$ columns. The **pseudoinverse** is $A^+ = V_r D^{-1} U_r^T$; $\hat{x} = A^+ b$ is the least-squares solution of $Ax = b$ with smallest norm.

**Condition number:** For invertible $A$, $\kappa(A) = \sigma_1/\sigma_n$ measures sensitivity of $Ax = b$ to perturbations.

# Chapter 10

# Single-Qubit Quantum Systems

*Sources: Eleanor Rieffel and Wolfgang Polak, Quantum Computing: A Gentle Introduction, Chapter 2; Sheldon Axler, Linear Algebra Done Right, §1A ($\mathbb{R}^n$ and $\mathbb{C}^n$). Optional: John Watrous, Basics of Quantum Information (video lectures).*

## 10.1 Complex numbers and $\mathbb{C}^n$ (Axler §1A)

**Complex numbers.** A complex number is $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$. Addition and multiplication follow the usual rules with $i^2 = -1$ (Axler defines these explicitly). The set $\mathbb{C}$ is a field: commutativity, associativity, identities, inverses, and distributivity hold (Axler 1.3). The **complex conjugate** is $\overline{a + bi} = a - bi$; **modulus** $|z| = \sqrt{z\overline{z}}$.

**Lists and $\mathbb{F}^n$.** A **list** of length $n$ is an ordered $n$-tuple $(z_1, \ldots, z_n)$. For $\mathbb{F}$ either $\mathbb{R}$ or $\mathbb{C}$, $\mathbb{F}^n$ is the set of all such lists with entries in $\mathbb{F}$. Addition and scalar multiplication are componentwise. Vectors in $\mathbb{C}^n$ are columns $(z_1, \ldots, z_n)^T$ when using matrix notation.

**Inner product on $\mathbb{C}^n$ (standard).** For $u = (w_1, \ldots, w_n)$ and $v = (z_1, \ldots, z_n)$ in $\mathbb{C}^n$,

$$\langle u, v \rangle = w_1\overline{z_1} + \cdots + w_n\overline{z_n}.$$

conjugate-linear in the *second* slot and linear in the first (matching Rieffel's bra/ket usage below). The **norm** is $\|v\| = \sqrt{\langle v, v \rangle}$.

**Scalars and $\mathbb{F}^n$ (Axler).** The letter $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$ (Axler 1.6). For fixed $n$, $\mathbb{F}^n$ is the set of all lists (ordered $n$-tuples) $(x_1, \ldots, x_n)$ with each $x_k \in \mathbb{F}$. Addition is coordinatewise; **scalar multiplication** is $\lambda(x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n)$ for $\lambda \in \mathbb{F}$. The zero vector is $0 = (0, \ldots, 0)$. These operations satisfy the vector-space axioms (proved in Axler from the field axioms for $\mathbb{F}$).

**Fields (Axler "Digression").** Real and complex numbers are examples of **fields**: sets with addition and multiplication satisfying commutativity, associativity, identities, inverses (except 0 has no multiplicative inverse), and distributivity. All results stated for $\mathbb{F}^n$ apply with $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

*Remark* 10.1.1 (Inner-product conventions). Axler develops abstract inner products in Chapter 6; his complex convention may place the complex conjugate on the *first* slot rather than the second. Rieffel–Polak and these notes use $\langle \phi | \psi \rangle$ with conjugate-linearity in the bra—keep one convention fixed in every calculation.

## 10.2 Qubit as $\mathbb{C}^2$ (Rieffel–Polak §2.2)

The **state space** of a two-level quantum system is a two-dimensional complex inner-product space. A **qubit** is a *unit* vector in $\mathbb{C}^2$, representing a physical state up to a **global phase** $e^{i\varphi}$ (Rieffel §2.5): $|\psi\rangle$ and $e^{i\varphi}|\psi\rangle$ are the same state.

Fix an orthonormal **computational** (or **standard**) basis $\{|0\rangle, |1\rangle\}$. Any (pure) state is

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle\,, \qquad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1.$$

The "two states" of a classical bit are $|0\rangle$ and $|1\rangle$; a qubit can be any superposition $a\,|0\rangle + b\,|1\rangle$ on the unit sphere in $\mathbb{C}^2$.

**Linear algebra vocabulary (Rieffel §2.2).** A ket is a **linear combination** of kets $|s_1\rangle, \ldots, |s_n\rangle$ if $|v\rangle = \sum_i a_i\,|s_i\rangle$ with $a_i \in \mathbb{C}$. The **span** of a set $S$ consists of all such combinations. A **basis** is a linearly independent spanning set; in finite dimensions every basis has the same length (**dimension**). For qubits, dim = 2; any two non-collinear kets form a basis, but quantum mechanics usually uses **orthonormal** bases: $\langle\beta_i|\beta_j\rangle = \delta_{ij}$.

## 10.3 Dirac notation: bras, kets, inner product

A **ket** $|\psi\rangle$ is a column vector. The Hermitian conjugate (row) is the **bra** $\langle\psi| = |\psi\rangle^\dagger$. The **inner product** $\langle\phi|\psi\rangle$ is linear in $|\psi\rangle$ and conjugate-linear in $\langle\phi|$. Rieffel lists the axioms: $\langle\psi|\psi\rangle \geq 0$ real; $\langle\phi|\psi\rangle = \overline{\langle\psi|\phi\rangle}$; and linearity in the second argument. Orthogonal means $\langle\phi|\psi\rangle = 0$; **orthonormal** bases satisfy $\langle\beta_i|\beta_j\rangle = \delta_{ij}$.

**Born rule:** A projective measurement in orthonormal basis $\{|b_0\rangle, |b_1\rangle\}$ gives outcome $k$ with probability

$$P(k) = |\,\langle b_k|\psi\rangle\,|^2,$$

and the post-measurement state is $|b_k\rangle$ (for an ideal projective measurement).

## 10.4 Photon polarization (Rieffel §2.1)

**Physical model.** Linear polarization of a photon is described by a unit vector in a real 2-dimensional plane (vertical $|\uparrow\rangle$, horizontal $|\rightarrow\rangle$, or an angle in between). More general polarization uses $\mathbb{C}^2$ (circular and elliptical polarization correspond to complex amplitudes).

**Polarizer interaction (Born rule).** A polarizer has a **preferred axis** (a unit ket $|u\rangle$). A photon in state $|\psi\rangle$ passes with probability $|\,\langle u|\psi\rangle\,|^2$ and is then measured to be in state $|u\rangle$; otherwise it is absorbed (orthogonal outcome).

**Three-polarizer "paradox" (Rieffel §2.1.1–2.1.2).** Two crossed polarizers (e.g. horizontal then vertical) transmit no light. Inserting a third polarizer at 45° between them allows some transmission: photons blocked classically can acquire amplitude in the vertical direction after passing the intermediate $|\diamondsuit\rangle$ (diagonal) polarizer, because the state is re-prepared in an intermediate basis before the final measurement.

## 10.5 Single-qubit measurement (Rieffel §2.3)

A **projective measurement** in an orthonormal basis $\{|b_0\rangle, |b_1\rangle\}$ yields outcome $k$ with probability $|\,\langle b_k|\psi\rangle\,|^2$, and (for ideal measurement) collapses the state to $|b_k\rangle$. The same physical qubit can be measured in the computational basis $\{|0\rangle, |1\rangle\}$, the Hadamard basis $\{|+\rangle, |-\rangle\}$, the circular basis $\{|i\rangle, |-i\rangle\}$, or any other orthonormal basis; probabilities depend on the chosen basis.

## 10.6 BB84 quantum key distribution (Rieffel §2.4)

The **BB84** protocol lets two parties share a secret random bit string using qubits sent in randomly chosen bases (computational or Hadamard). Comparing a subset of bases and outcomes over a public channel reveals eavesdropping: an interceptor who measures in the wrong basis disturbs

the state. This uses the fact that non-orthogonal states cannot be distinguished reliably without disturbing them—a theme beyond the SP5 exercise list but central to Rieffel's §2.4.

## 10.7 State space: global phase, $\mathbb{C}\mathrm{P}^1$, Bloch sphere (Rieffel §2.5)

Unit vectors $|\psi\rangle$ and $e^{i\varphi}|\psi\rangle$ describe the **same** physical state (**global phase**). Observable phases are **relative** (e.g. the phase between amplitudes of $|0\rangle$ and $|1\rangle$ in a fixed basis). The space of distinct pure states of one qubit is the complex projective line $\mathbb{C}\mathrm{P}^1$ (quotient of the unit sphere in $\mathbb{C}^2$ by the global-phase equivalence). The **Bloch sphere** parametrizes $\mathbb{C}\mathrm{P}^1$ by a point on the ordinary sphere $S^2$, giving a geometric picture of qubit state space (Rieffel §2.5.2).

## 10.8 Named bases for one qubit

**Computational:** $\{|0\rangle, |1\rangle\}$.

**Hadamard:**

$$|+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad |-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

**Circular ($y$-axis) basis:**

$$|i\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle), \qquad |-i\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle).$$

(Some texts denote circular polarization with separate symbols; always match the problem's definitions.)

A state is a **superposition** with respect to a chosen orthonormal basis if at least two basis coefficients are nonzero. The same physical state may be a superposition in one basis and a basis vector in another (e.g. $|+\rangle$ in $\{|0\rangle, |1\rangle\}$ vs. $\{|+\rangle, |-\rangle\}$).

## 10.9 Useful Algebra for SP5

**Same-state test.** Two nonzero kets $|\psi\rangle$ and $|\phi\rangle$ represent the same physical state if and only if

$$|\psi\rangle = e^{i\theta}|\phi\rangle$$

for some real $\theta$. In coordinates, this means the coefficient vectors are scalar multiples of each other by a complex number of modulus 1.

**Basis conversions.** The identities used most often in the assigned problems are

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle), \qquad |1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle),$$

and

$$|0\rangle = \frac{1}{\sqrt{2}}(|i\rangle + |-i\rangle), \qquad |1\rangle = \frac{1}{i\sqrt{2}}(|i\rangle - |-i\rangle).$$

These let you rewrite a state in whichever basis the problem asks about.

**Superposition test.** In a basis $\{|b_0\rangle, |b_1\rangle\}$, a state

$$|\psi\rangle = \alpha\,|b_0\rangle + \beta\,|b_1\rangle$$

is a superposition precisely when *both* coefficients are nonzero.

**Measurement workflow.** To measure $|\psi\rangle$ in orthonormal basis $\{|b_0\rangle, |b_1\rangle\}$:

1. Compute the amplitudes $\langle b_0|\psi\rangle$ and $\langle b_1|\psi\rangle$.

2. Square magnitudes to get probabilities:

$$P(b_0) = |\langle b_0|\psi\rangle|^2, \qquad P(b_1) = |\langle b_1|\psi\rangle|^2.$$

3. Check that the probabilities add to 1.

*Remark* 10.9.1 (Finding a basis in which a state is not a superposition). If $|\psi\rangle$ is normalized, then any orthonormal basis of the form $\{|\psi\rangle, |\psi^\perp\rangle\}$ makes $|\psi\rangle$ a basis vector rather than a superposition. In $\mathbb{C}^2$, once $|\psi\rangle$ is known, the orthogonal complement is one-dimensional, so $|\psi^\perp\rangle$ is unique up to phase.

## 10.10 Suggested Problems 5 (course alignment)

**Suggested reading:** Axler, *Linear Algebra Done Right*, §1A; Watrous, *Basics of Quantum Information* (videos); Rieffel–Polak, §2.
**Quiz-level:** Rieffel–Polak Exercises 2.2, 2.3, 2.4, 2.6.
**Exam-level:** N/A (similar difficulty to quiz-level).
**Code:** `ex9_quantum_qubit.m` (Octave checks for inner products and Born probabilities).

*Remark* 10.10.1 (What these notes cover). The sections above follow the *logical order* of ideas in Axler §1A and Rieffel §2; they are not a substitute for reading the books (examples, figures, proofs, and all exercises). **SP5 exercises solved in the solutions PDF:** Rieffel 2.2, 2.3, 2.4, 2.6 only (per the course handout). Rieffel's other Chapter 2 exercises (e.g. 2.1, 2.5, 2.7–2.14) are not assigned on SP5 but appear in the textbook for extra practice.

# Chapter 11

# Image Compression

*Sources: Justin Wyss-Gallifent, Image Compression; Lay & McDonald, Linear Algebra and Its Applications, §7.5.*

*Remark* 11.0.1 (Suggested Problems 4). **Quiz:** Justin 10.1–10.3. **Exam:** Justin 10.6. (Lay §7.5.1–6: practice for PCA/covariance.)

## 11.1 Image Representation

A grayscale image displayed pixel-by-pixel is stored as a matrix. Common conventions:

- **Color:** Red, Green, Blue each 0–255 ($256^3$ colors).
- **Grayscale (integer):** Single value 0–255 (0 = black, 255 = white).
- **Grayscale (real):** Single value in $[0, 1]$ (0 = black, 1 = white).

We use the third option (real values in $[0, 1]$) for mathematical convenience.

## 11.2 Image Compression via SVD

If a matrix $A$ represents an image, then a low-rank approximation $A' = U\Sigma' V^T$ (with some singular values zeroed) approximates the image while preserving a fraction of the variance.

**Handling out-of-range values:** After reconstruction, entries may fall outside $[0, 1]$. Do *not* rescale (that would distort meaning). Instead: treat values $> 1$ as white and values $< 0$ as black.

**Effect:** As fewer singular values are kept, the image becomes more uniform while retaining as much variance as possible. Rank-1 images are a single vertical vector (shades of grey) scaled across columns.

If

$$A = U\Sigma V^T = \sum_{i=1}^{r} \sigma_i u_i v_i^T,$$

then the rank-$k$ compressed image is

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T.$$

This is the best rank-$k$ approximation to $A$ in Frobenius norm.

## 11.3 Image Quality

**Definition 11.3.1** (Image Quality)**.** The **image quality** (proportion of variance preserved) is

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_{\min(m,n)}^2}.$$

A typical target is $\geq 99.5\%$ variance for a reasonable-looking picture.

*Remark* 11.3.1. The singular values should be ordered from largest to smallest. In image problems, this means we always keep the *first* $k$ singular values, never an arbitrary collection.

## 11.4 Data Savings

For an $n \times n$ grayscale image:

- **Original:** $n^2$ values (one per pixel).
- **SVD with $k$ singular values:** We store $\bar{u}_1, \ldots, \bar{u}_k$ (each $n$ values) and the coefficients for each column (each column needs $k$ coefficients, $n$ columns $\Rightarrow$ $kn$ values). Total: $kn + kn = 2kn$ values.

**Definition 11.4.1** (Compression Ratio)**.**

$$\frac{2kn}{n^2} = \frac{2k}{n}.$$

**Compression is beneficial** when $2kn < n^2$, i.e., $k < n/2$.

For a general $m \times n$ image, storing the first $k$ singular triplets requires

$$km \quad \text{(left singular vectors)} \qquad + \qquad kn \quad \text{(right singular vectors)} \qquad + \qquad k \quad \text{(singular values)}.$$

So the exact storage count is

$$k(m + n + 1),$$

and a standard large-scale approximation is $k(m + n)$ when the extra $k$ singular values are negligible compared to $km$ and $kn$.

**Definition 11.4.2** (General Compression Ratio)**.** For an $m \times n$ image compressed with $k$ singular values, an exact compression ratio is

$$\frac{k(m + n + 1)}{mn}.$$

Ignoring the comparatively small $k$ singular values gives the common approximation

$$\frac{k(m + n)}{mn} = k\left(\frac{1}{m} + \frac{1}{n}\right).$$

**Compression is beneficial** when the stored data are fewer than the original $mn$ entries, i.e.

$$k(m + n + 1) < mn$$

or approximately

$$k < \frac{mn}{m + n}.$$

## 11.5 Problem-Solving Workflow

For the image-compression exercises, the workflow is always:

1. Translate the image into a matrix $A$ with entries in $[0, 1]$.
2. Compute an SVD $A = U\Sigma V^T$.
3. Read the singular values from the diagonal of $\Sigma$.
4. Form $A_k$ by zeroing out all but the first $k$ singular values.
5. Compute the quality
$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_r^2}.$$
6. Compare the storage for the compressed representation with the original matrix.

For small hand-drawn images, the main ideas are:

- identify the matrix correctly,
- keep track of how the picture changes as $k$ decreases,
- report quality and compression clearly.

## 11.6 Principal Component Analysis (Lay-McDonald §7.5)

For multivariate data $X_1, \ldots, X_N$ in $\mathbb{R}^p$, the **sample mean** is $M = \frac{1}{N}(X_1 + \cdots + X_N)$. Put data in **mean-deviation form**: $\hat{X}_k = X_k - M$. The **sample covariance matrix** is

$$S = \frac{1}{N-1} BB^T, \quad B = [\hat{X}_1 \ \cdots \ \hat{X}_N].$$

$S$ is symmetric and positive semidefinite. The **principal components** are the unit eigenvectors of $S$ ordered by decreasing eigenvalue. The first principal component $u_1$ gives the direction of maximum variance; $y_1 = u_1^T X$ has the largest variance among unit linear combinations. The fraction of total variance in the first component is $\lambda_1 / \operatorname{tr}(S)$.